Project Report

Project Title

Integrated Procure-to-Pay (P2P) Analytics with FI Integration in SAP for Data-Driven Procurement Control

Candidate Details

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College Batch and Program	**2023-27** **BTech CSE**
SAP Program	**2026 SAP Data Analytics Engineer**

Submission Details

Domain	SAP Data Analytics Engineer
Project Type	Individual Capstone Project
Selected Scenario	Procure-to-Pay (P2P) - SAP MM with FI integration
Date	April 2026

1. Problem Statement

Many organizations face delayed procurement visibility, uncontrolled vendor spending, and poor tracking of purchase lifecycle bottlenecks. In traditional reporting, procurement teams and finance teams work with delayed and disconnected data. As a result, management cannot quickly answer key questions: -

- Which vendors have high lead times?
- Where are purchase requisitions pending conversion?
- Which purchase orders are delayed at goods receipt stage?
- How much liability is open due to pending invoice posting and payment?

The absence of a structured end-to-end analytics model across SAP MM and FI causes inefficient sourcing decisions, delayed month-end closure, and unnecessary working capital blocking.

2. Proposed Solution

This project builds an end-to-end P2P process blueprint in SAP with analytics-ready checkpoints.

2.1 Process Scope

- Master Data Setup: Company code, plant, storage location, purchasing organization, purchasing group, material master, vendor master
- Transactional Flow: PR -> RFQ/Quotation (optional) -> PO -> Goods Receipt -> Invoice Verification -> Outgoing Payment
- FI Integration: Automatic account postings for GR/IR, vendor liabilities, and clearing entries
- Analytics Layer: KPI model and query views for procurement and finance performance monitoring

2.2 Functional Modules Used

- SAP MM: Procurement lifecycle
- SAP FI: Invoice accounting and vendor payment integration
- Reporting Layer: SQL/HANA-style semantic view design (tool-agnostic, dashboard-ready)

3. Business Scenario (Fictitious Company)

3.1 Organization Profile

- Company Name: Nova Retail Manufacturing Pvt. Ltd.
- Industry: Retail + Light Assembly
- Headquarters: Bhubaneswar

- Procurement Objective: Reduce procurement cycle time and improve invoice payment compliance

3.2 Organizational Structure

- Company Code: NR01
- Plant: PL01
- Storage Location: SL01
- Purchasing Organization: PO01
- Purchasing Group: PG1

3.3 Organizational Mapping Table

Organization Element	Code	Purpose
Company Code	NR01	Legal and accounting reporting unit
Plant	PL01	Operational procurement and receiving location
Storage Location	SL01	Material stock storage point
Purchasing Organization	PO01	Central procurement control unit
Purchasing Group	PG1	Buyer/team responsible for purchasing activity

3.4 Procurement Use Case

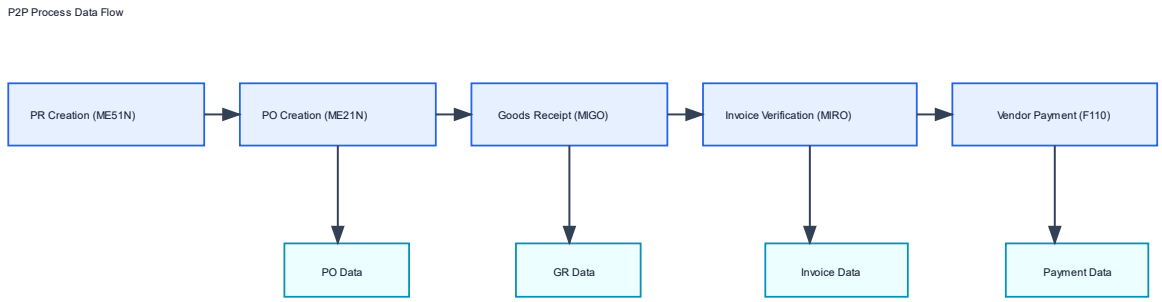
The company procures packaging material and spare components from multiple vendors. Management needs a dashboard that tracks: - PR to PO conversion time - PO to GR lead time - GR to Invoice delay - Invoice to Payment cycle - Vendor-wise spend and rejection rate

4. End-to-End Process and Data Flow

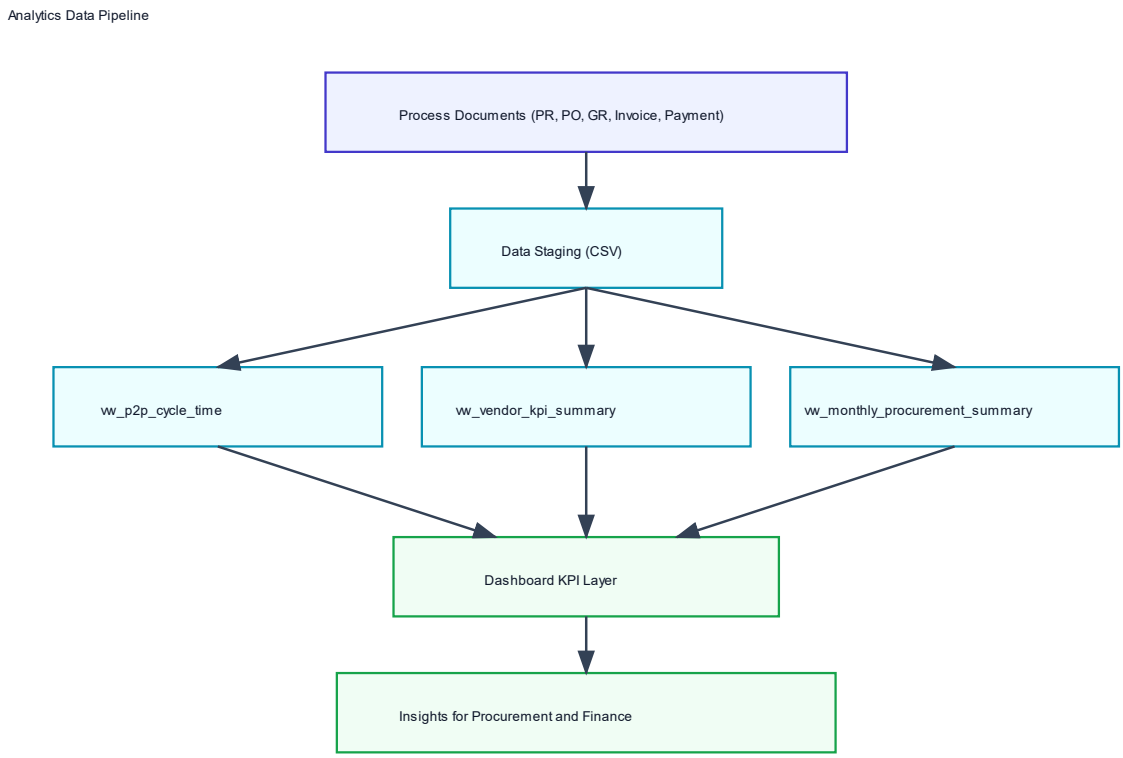
4.1 Transaction Flow Table

Stage	SAP T-Code	Primary Input	Output Document	Analytics Checkpoint
Purchase Requisition	ME51N	Material, quantity, plant, date	PR Number	PR creation timestamp
Purchase Order	ME21N	PR reference, vendor, price	PO Number	PR-to-PO cycle time
Goods Receipt	MIGO	PO reference, received quantity	GR Document	PO-to-GR lead time
Invoice Verification	MIRO	PO/GR reference, invoice value	Invoice Document	GR-to-Invoice lag
Payment	F110	Due invoice, payment method	Payment Document	Invoice-to-Payment lag

4.2 P2P Process Data Flow Diagram



4.3 Analytics Data Pipeline Diagram



5. Key Features Implemented

1. End-to-end P2P process documentation with transaction-level mapping
2. MM-FI integration checkpoints and posting logic
3. Master data and transaction templates for repeatable execution and validation
4. KPI dictionary for operational and financial analytics

5. SQL analytical model to generate dashboard datasets
6. Test case set for validating process integrity and analytics accuracy

6. KPI Framework and Reference Tables

6.1 KPI Definition Table

KPI Name	Formula	Unit	Business Use
PR to PO Conversion Time	po_date - pr_date	Days	Measures sourcing responsiveness
PO to GR Lead Time	gr_date - po_date	Days	Measures supplier delivery performance
GR to Invoice Lag	invoice_date - gr_date	Days	Measures invoice processing efficiency
Invoice to Payment Lag	payment_date - invoice_date	Days	Measures payables cycle efficiency
Vendor On-Time Rate	on_time_receipts / total_receipts	Percentage	Vendor reliability tracking
Spend by Vendor	Sum(po_value)	Currency	Vendor concentration and cost visibility
Price Variance Percent	variance_amount / invoice_amount * 100	Percentage	Price and invoice control

6.2 Sample Analytical Output Table

Vendor	Transactions	Total PO Value (INR)	Avg PR->PO Days	Avg PO->GR Days	On-Time Rate %
V1001	3	297000	2.0	5.3	66.7
V1002	2	158000	3.0	8.0	0.0
V1003	1	210000	2.0	9.0	0.0
V1004	1	98000	2.0	2.0	100.0
V1005	1	155000	2.0	3.0	100.0

7. Tech Stack

- SAP ERP process blueprint (MM + FI functional flow)
- SQL (HANA-style analytical views)
- Spreadsheet/CSV for sample data staging
- Optional BI Layer: Power BI / SAP Analytics Cloud for dashboard visualization
- Documentation: Markdown source converted to PDF

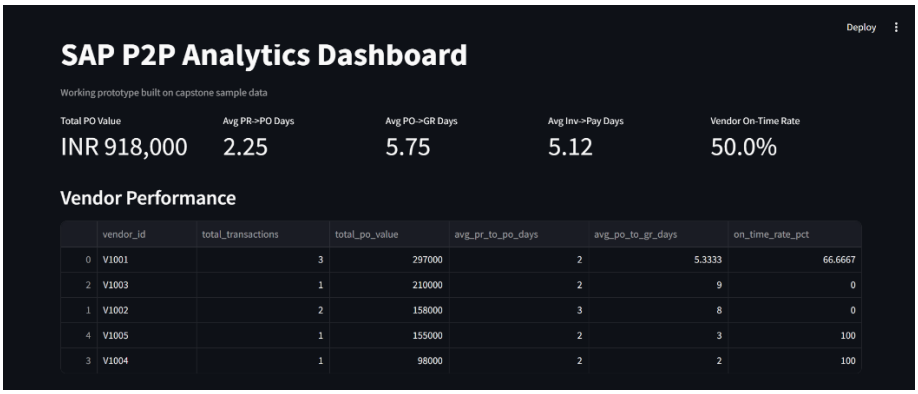
8. Unique Points of the Project

1. Combines process execution and analytics engineering instead of only transaction walkthrough

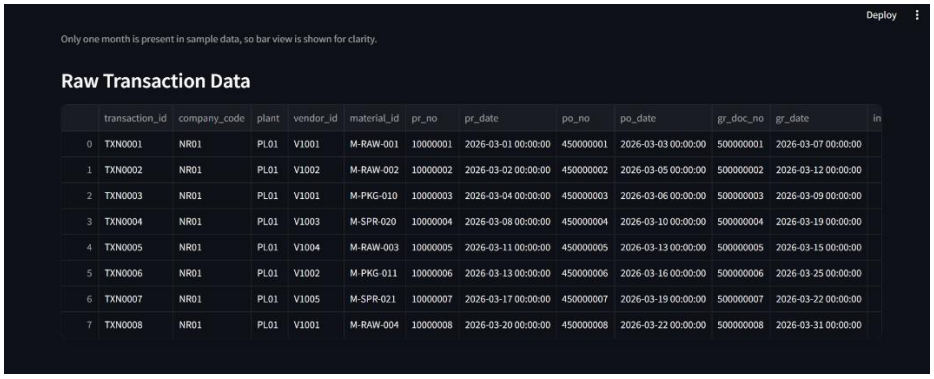
2. Maps each procurement stage to measurable KPIs and business impact
3. Includes FI financial checkpoints for liability and payment visibility
4. Provides test cases and validation method for quality assurance
5. Structured for quick demo and viva explanation

9.Screenshots and artifacts

The dashboard is generated from the file “p2p_sample_transactions.csv,” and the KPI logic follows the same definitions used in the report.



	vendor_id	total_transactions	total_po_value	avg_pr_to_po_days	avg_po_to_gr_days	on_time_rate_pct
0	V1001	3	297000	2	5.3333	66.6667
2	V1003	1	210000	2	9	0
1	V1002	2	158000	3	8	0
4	V1005	1	155000	2	3	100
3	V1004	1	98000	2	2	100



	transaction_id	company_code	plant	vendor_id	material_id	pr_no	pr_date	po_no	po_date	gr_doc_no	gr_date	in
0	TXN0001	NR01	PL01	V1001	M-RAW-001	10000001	2026-03-01 00:00:00	450000001	2026-03-03 00:00:00	500000001	2026-03-07 00:00:00	
1	TXN0002	NR01	PL01	V1002	M-RAW-002	10000002	2026-03-02 00:00:00	450000002	2026-03-05 00:00:00	500000002	2026-03-12 00:00:00	
2	TXN0003	NR01	PL01	V1001	M-PKG-010	10000003	2026-03-04 00:00:00	450000003	2026-03-06 00:00:00	500000003	2026-03-09 00:00:00	
3	TXN0004	NR01	PL01	V1003	M-SPR-020	10000004	2026-03-08 00:00:00	450000004	2026-03-10 00:00:00	500000004	2026-03-19 00:00:00	
4	TXN0005	NR01	PL01	V1004	M-RAW-003	10000005	2026-03-11 00:00:00	450000005	2026-03-13 00:00:00	500000005	2026-03-15 00:00:00	
5	TXN0006	NR01	PL01	V1002	M-PKG-011	10000006	2026-03-13 00:00:00	450000006	2026-03-16 00:00:00	500000006	2026-03-25 00:00:00	
6	TXN0007	NR01	PL01	V1005	M-SPR-021	10000007	2026-03-17 00:00:00	450000007	2026-03-19 00:00:00	500000007	2026-03-22 00:00:00	
7	TXN0008	NR01	PL01	V1001	M-RAW-004	10000008	2026-03-20 00:00:00	450000008	2026-03-22 00:00:00	500000008	2026-03-31 00:00:00	

10. Future Improvements

1. Add predictive vendor delay model using historical lead-time patterns
2. Add anomaly detection for invoice price variance
3. Integrate with real-time APIs or SAP BW extractors
4. Extend flow with inventory optimization KPIs
5. Build role-based dashboard views for buyer, finance manager, and leadership

11. Conclusion

This capstone demonstrates how SAP transaction flows can be converted into actionable analytics. The project closes the gap between process execution design and business intelligence by delivering a measurable, dashboard-ready P2P model. This improves procurement governance, strengthens finance visibility, and supports data-driven decision-making in enterprise operations.